1. Information about the indicator	
Goal	Taking urgent measures to combat climate change and its consequences
Task	Improve education, information dissemination, and the capacity of people and institutions to mitigate, adapt to, and prevent climate change.
The indicator	Total annual greenhouse gas emissions
2. Information about the organization	
Organization	Ministry of Ecology and Natural Resources of the Republic of Kazakhstan
3. Definitions and concepts	
Definition	The total annual volume of greenhouse gas emissions is the total volume of anthropogenic greenhouse gas emissions generated in the territory of the Republic of Kazakhstan for a calendar year, expressed in carbon dioxide equivalent (Co_2-eq.)
Basic concepts:	Greenhouse gases (GHGs) are gases that trap thermal radiation in the atmosphere and affect climate change (co_2, ch_4, N_2O, HFCs, PFCs, sf_6, etc.). The CO_2 equivalent is a single measure for calculating emissions of various GHGs, taking into account their global warming potential. The UNFCCC (United Nations Framework Convention on Climate Change) is an international agreement that sets out the obligations of States to account for and reduce greenhouse gas emissions and submit national reports. The Paris Agreement is an international treaty adopted within the framework of the UNFCCC aimed at limiting global temperature growth and reducing greenhouse gas emissions. The IPCC (Intergovernmental Panel on Climate Change) is an international scientific body that develops guidelines and methodologies for estimating and inventorying greenhouse gas emissions and removals. The National GHG Inventory is the official system for accounting for greenhouse gas emissions and removals in accordance with the requirements of the UNFCCC and the IPCC methodology.
Justification and interpretation	The indicator reflects the cumulative level of anthropogenic greenhouse gas emissions and is used to assess the fulfillment of the Republic of Kazakhstan's international obligations under the UNFCCC and the Paris Agreement. The values of the indicator are used in the formation of climate policy, monitoring of nationally determined contributions and planning measures for the transition to low-carbon development.
4. Data sources and collection methods	
Data sources	Statistical data
Data collection methods	Data collection is carried out as part of the annual preparation of the national greenhouse gas inventory based on: • Official government statistics on economic sectors; • departmental data of government agencies
Unit of measure ment	a ton of CO2 eq.
Periodicity	annually
5. Calculation method and other methodological considerations	
Calculation method:	The calculation of the total annual greenhouse gas emissions is

	carried out within the framework of the national GHG inventory by sectors defined by the IPCC.: • Energy • Industrial processes and product use • Agriculture • Land use, land-use change and forestry (LULUCF) • Waste The methodology is based on the 2006 IPCC Guidelines.
Comments and restrictions:	
6. Data availability and disaggregation	
Data availability and gaps:	The data for determining the indicator is available within the framework of the national greenhouse gas inventory system of the Republic of Kazakhstan and is generated on an annual basis. However, there may be some gaps related to the limited detail of statistical information on certain categories of emission sources.
The level of disaggregation:	The data is presented at the national level and can be disaggregated by economic sector according to the IPCC classification.
7. Comparability with international data/standards	The indicator is fully comparable with international reporting, as it is formed in accordance with: • IPCC methodology; • UNFCCC requirements for national inventory reports; • Transparency mechanisms of the Paris Agreement.
8. References and documentation	https://www.ipcc-nggip.iges.or.jp/public/2006gl/index.html https://unfccc.int/ghg-inventories-annex-i-parties/2025